(i) $a^2 = b^2 + c^2 - 2bc\cos A$ says that the square of a side is the sum of squares of other two sides diminished by twice the product of those two sides and the cosine of the included angle. Also one formula will give the other formula by cycling through the letters $a, b, c$.

(ii) The Laws of Cosine for right triangles reduce to Pythagorean theorem. Thus, the Law of cosines can be viewed as a generalisation of Pythagorean theorem.

(iii) The advantage of using law of cosines over law of sines is that unlike the sine function, the cosine function distinguishes between acute and obtuse angles. If cosine of an angle is positive, then it is acute. Otherwise, it is obtuse.

(iv) **The Law of Cosines says** : The direct route is the shortest. Let us explain this. In a $\triangle ABC$, $c^2 = a^2 + b^2 - 2ab\cos C$. Since $-\cos C < 1$ we have $c^2 < a^2 + b^2 + 2ab$. Thus, we have $c < a + b$. Hence, In $\triangle ABC$, we have $a < b + c,\ \ b < c + a,\ \ c < a + b$

(v) When using the law of cosines, it is always best to find the measure of the largest unknown angle first, since this will give us the obtuse angle of the triangle if there is one such angle.

### 3.7.3 Projection Formula

**Theorem 3.4:** In a $\triangle ABC$, we have

(i) $a = b\cos C + c\cos B$, (ii) $b = c\cos A + a\cos C$, (iii) $c = a\cos B + b\cos A$

*Proof.* In $\triangle ABC$, we have $a = BC$. Draw $AD \perp BC$.

$$\begin{aligned} a &= BC = BD + DC \\ &= \frac{BD}{AB}AB + \frac{DC}{AC}AC \\ &= (\cos B)c + (\cos C)b \\ a &= b\cos C + c\cos B \end{aligned}$$

Similarly, one can prove the other two projection formulas.

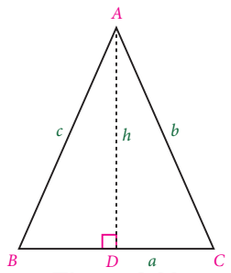


**Figure 3.20**

$a = b\cos C + c\cos B$ says that $a =$ projection of $b$ on $a +$ projection of $c$ on $a$
Thus, a side of triangle is equal to sum of the projections of other two sides on it.

### 3.7.4 Area of the Triangle

We shall use some elements of an oblique triangle and the sine function to find the area of the triangle. Recall that area formula for $\triangle ABC$ is $\frac{1}{2}bh$ where $b$ is the base and $h$ is the height. For oblique triangle, we must find $h$ before using the area formula.

**Theorem 3.5:** In $\triangle ABC$, area of the triangle is

$$\triangle = \frac{1}{2}\,ab\sin C = \frac{1}{2}\,bc\sin A = \frac{1}{2}\,ac\sin B$$

*Proof.* In $\triangle ABC$, draw $AD \perp BC$

In $\triangle ADC$, $\dfrac{AD}{AC} = \sin C \Rightarrow AD = b\sin C$

Thus, $\triangle = \dfrac{1}{2} \times$ base $\times$ height $= \dfrac{1}{2}\,ab\sin C.$

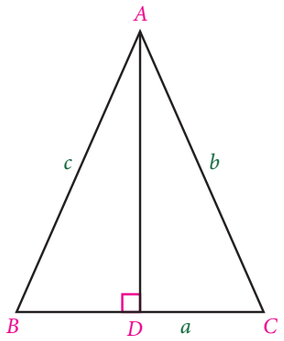


**Figure 3.21**

Similarly we can derive the other two results.